

Module 2

Engineering Design Approaches

What is design?

- To design is to pull together something new or to arrange existing things in a new way to satisfy a recognized need of society.
- The ability to design is both a science and an art.
- Design should not be confused with discovery. We can discover what has already existed but has not been known before, but a design is the product of planning and work.
- Design may or may not involve invention.

Design: Analysis and Synthesis

- Good design requires both analysis and synthesis.
- Typically we approach complex problems like design by decomposing the problem into manageable parts. Because we need to understand how the part will perform in service, we must be able to calculate as much about the part's expected behavior as possible before it exists in physical form by using the appropriate disciplines of science and engineering science and the necessary computational tools.
- Synthesis involves the identification of the design elements that will comprise the product, its decomposition into parts, and the combination of the part solutions into a total workable system.

The Four C's of Design

□ Creativity

- ✓ Requires creation of something that has not existed before or has not existed in the designer's mind before.

□ Complexity

- ✓ Requires decisions on many variables and parameters.

□ Choice

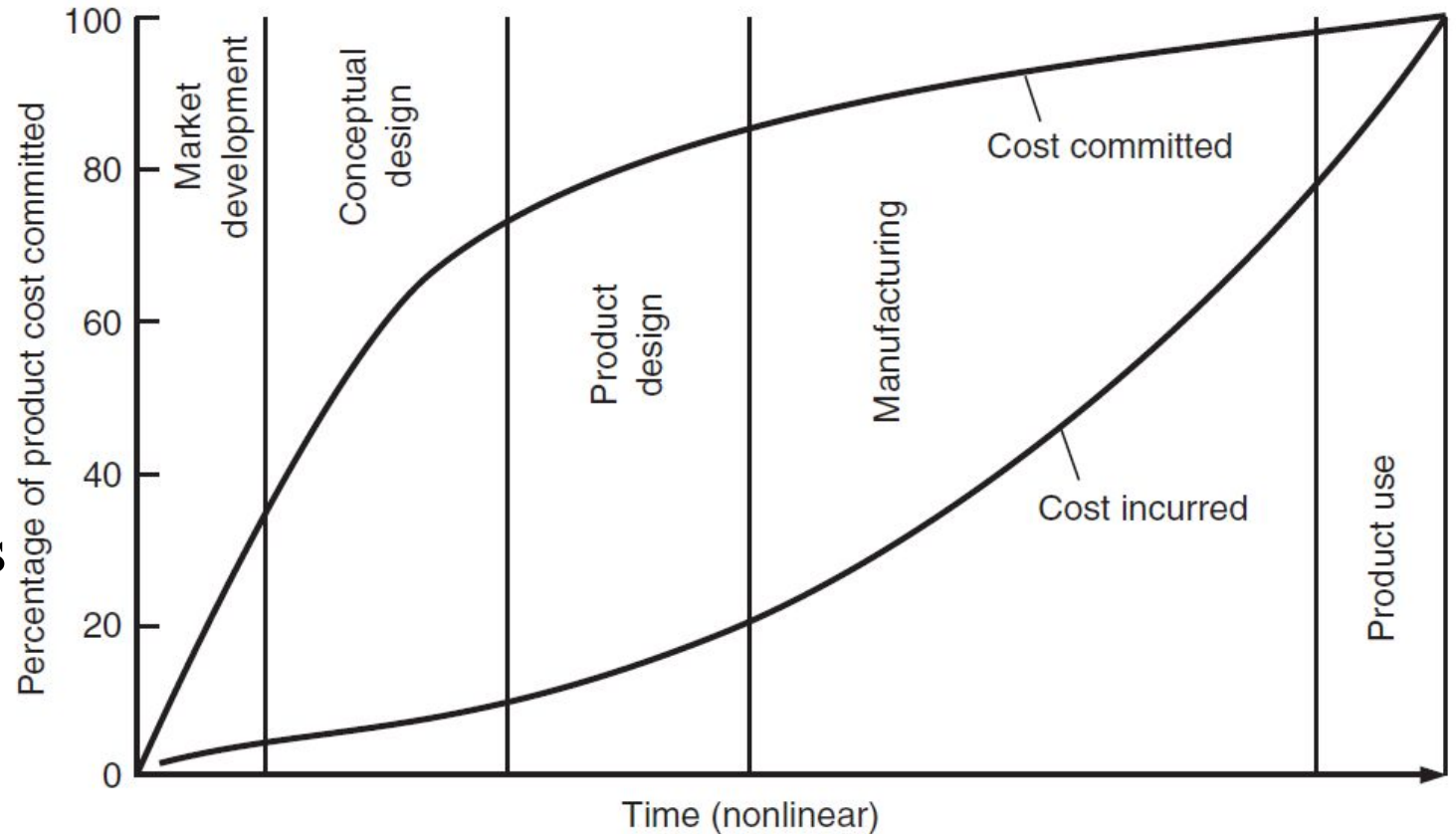
- ✓ Requires making choices between many possible solutions at all levels, from basic concepts to the smallest detail of shape.

□ Compromise

- ✓ Requires balancing multiple and sometimes conflicting requirements.

Importance of the Engineering Design Process

- Cost of design
 - Product quality
 - Product cycle time
- ❖ The design process should be conducted so as to develop quality, cost-competitive products in the shortest time possible.



Types of Designs

❑ **Original design**

This form of design is at the top of the hierarchy. It employs an original, innovative concept to achieve a need. Sometimes but rarely, the need itself may be original.

A truly original design involve invention. Example is steam engine.

❑ **Adaptive design**

This form of design occurs when the design team adapts a known solution to satisfy a different need to produce a novel application.

❑ **Redesign**

To modify shape of existing design, working principle is same.

❑ **Selection design**

To modify the parts used in machine

A Problem-Solving Methodology

❖ Designing can be approached as a problem to be solved. A problem-solving methodology that is useful in design consists of the following steps

- Definition of the problem
- Gathering of information
- Generation of alternative solutions
- Evaluation of alternatives and decision making
- Communication of the results

Definition of the problem

- The most critical step in the solution of a problem is the *problem definition* or formulation.
- The formulation of the problem should start by writing down a problem statement. This document should express as specifically as possible what the problem is.
- It should include objectives and goals, the current state of affairs and the desired state, any constraints placed on solution of the problem, and the definition of any special technical terms
- Problem definition often needs analysis

Gathering of information

- Textbooks and articles published in the scholarly technical journals usually are of lesser importance. The need often is for more specific and current information than is provided by those sources.
- Technical reports published as a result of government-sponsored R&D, company reports, trade journals, patents, catalogs, and handbooks and literature published by vendors and suppliers of material and equipment are important sources of information.
- The Internet is a very useful resource
- Discussions with in-house experts (often in the corporate R&D center) and outside consultants may prove helpful

The following are some of the questions concerned with obtaining information

- What do I need to find out?
- Where can I find it and how can I get it?
- How credible and accurate is the information?
- How should the information be interpreted for my specific need?
- When do I have enough information?
- What decisions result from the information?

Generation of Alternative Solutions

- Generating alternative solutions or design concepts involves the use of creativity stimulation methods, the application of physical principles and qualitative reasoning, and the ability to find and use information.
- The ability to generate high-quality alternative solutions is vital to a successful design.

Evaluation of Alternatives and Decision Making

- Engineering analysis procedures
- Simulation of performance with computer models
- Mathematical checks and engineering-sense checks
- Robust design that is resistant to environmental influences (water vapor, temperature, vibration, etc.)

Communications of the results

- Satisfy the needs of a customer or client
- Detailed engineering drawings, computer programs, 3-D computer models, and working models

Design-Social and Professional context

Design-Social and Professional context

❖ *Social - Design to solve social issue or need*

- ✓ Ex- Devices for Handicapped
- ✓ Better housing
- ✓ Better sanitation
- ✓ Clean drinking water
- ✓ Abundant food
- ✓ Educational help

Note: Unsocial designs degrade the quality of life

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❖ *Professional - Design focused on comfortability, ease to access or technological advancement*

- ✓ Ex- E-Newspaper
- ✓ Business class in aircraft
- ✓ Electric bicycles
- ✓ Wireless charger
- ✓ Headphones/wireless headphones
- ✓ Sensors watertap

Different Phases of Design Process

Three Major Phases

1. Conceptual Design
2. Embodiment Design
3. Detailed Design

Phase I. Conceptual Design

- Conceptual design is the process by which the design is initiated, carried to the point of creating a number of possible solutions, and narrowed down to a single best concept.
- It is sometimes called the feasibility study

❖ *Identification of customer needs*

The goal of this activity is to completely understand the customers needs and to communicate them to the design team.

❖ *Problem definition*

- Quality function deployment (QFD) is a valuable tool for linking customer needs with design requirements.
- A detailed listing of the product requirements is called a product design specification (PDS).

❖ *Gathering information*

Engineering design presents special requirements over engineering research in the need to acquire a broad spectrum of information

❖ *Conceptualization*

Concept generation involves creating a broad set of concepts that potentially satisfy the problem statement. Team-based creativity methods, combined with efficient information gathering, are the key activities

❖ *Concept selection*

Evaluation of the design concepts, modifying and evolving into a single preferred concept, are the activities in this step

❖ *Refinement of the PDS*

The product design specification is revisited after the concept has been selected. The design team must commit to achieving certain critical values of design parameters, usually called critical-to-quality (CTQ) parameters

❖ *Design review*

- Before committing funds to move to the next design phase, a design review will be held.
- The design review will assure that the design is physically realizable and that it is economically worthwhile.
- It will also look at a detailed product-development schedule.
- This is needed to devise a strategy to minimize product cycle time and to identify the resources in people, equipment, and money needed to complete the project.

Phase II. Embodiment Design

- Structured development of the design concept occurs in this engineering design phase.
- It is the place where flesh is placed on the skeleton of the design concept.
- It is in this design phase that decisions are made on strength, material selection, size, shape, and spatial compatibility.
- Beyond this design phase, major changes become very expensive.

❖ *Determining product architecture*

Product architecture is concerned with dividing the overall design system into subsystems or modules. In this step we decide how the physical components of the design are to be arranged and combined to carry out the functional duties of the design

❖ *Configuration design of parts and components*

- Parts are made up of features like holes, ribs, splines, and curves
- Preliminary selection of materials, modeling and size of parts

❖ *Parametric design of parts*

- Creating a robust design, and selection of final dimensions/parameters and tolerances
- Parametric design starts with information on the configuration of the part and aims to establish its exact dimensions and tolerances
- Final decisions on the material and manufacturing processes are also established if this has not been done previously
- Robustness refers to how consistently a component performs under variable conditions in its service environment
- Parametric design also deals with determining the aspects of the design that could lead to failure
- Reduction of manufacturing cost

Phase III. Detail Design

- In this phase the design is brought to the stage of a complete engineering description of a tested and producible product.
- Missing information is added on the arrangement, form, dimensions, tolerances, surface properties, materials, and manufacturing processes of each part
- Detailed engineering drawings suitable for manufacturing. Routinely these are computer-generated drawings, and they often include three-dimensional CAD models
- Verification testing of prototypes is successfully completed and verification data is submitted. All critical-to-quality parameters are confirmed to be under control.
- Creation of final drawings and/or specifications
- Assembly drawings and assembly instructions also will be completed. The bill of materials for all assemblies will be completed

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- A detailed product specification, updated with all the changes made since the conceptual design phase, will be prepared.
- Decisions on whether to make each part internally or to buy from an external supplier will be made
- With the preceding information, a detailed cost estimate for the product will be carried out
- Finally, detail design concludes with a design review before the decision is made to pass the design information on to manufacturing

Identification and Specifications

Need Identification and Specification

- For design process, we require need and specification
- Needs arise from a number of sources including:
 - Incremental improvements to existing designs
 - Replacements for existing designs
- Categories of Needs:
 - (1) **Minimum needs** ==> Are the new features that the customers consider important
Ex. For car this would mean seating and dashboard electronics
 - (2) **Assumed needs** ==> It includes basic operation, safety and reliability.
Ex. For a car this would mean the ability to drive
 - (3) **Unrecognized needs** ==> Ex. . For a car this would mean sounds, AC, voltage requirement

- Design objectives and functional requirements are what the design should strive for. It is best to define the ideal and acceptable values
- Examples of well-defined objectives include the following:
 - ❖ Cost between \$100.00 and \$120.00
 - ❖ Provide a maximum power from 50 to 75 KW
 - ❖ A maximum speed of 100 to 150 m/s
 - ❖ Heating time between 40 and 60 s
- Design constraints are firm limits that must be met for the design to be acceptable. Examples of these are listed below
 - ❖ Must be delivered by June 15
 - ❖ Surface temperature below 50°C
 - ❖ Cycle times below 3 seconds
 - ❖ Power levels above 5 W

Specifications

- A specific description of design and materials used to make something
- They are set of instructions or requirements given in codes and standards
- Must meet requirements of customers
- Example-Company provided specification, Product design specifications

Requirements of specifications

- Specifications provide specific/ additional requirements for materials, components based on codes and standards
- Often generated by companies to address additional requirements applicable to a specific products or application

Conversion of needs to specification

When converting needs to specification, approaches to consider include the following

- Find similar designs from the past, present, or future.
- Examine each of the objectives and constraints and find methods for implementation.
- Identify components
- Meeting with customers, users, or others
- Perform benchmarking with competitive designs
- Identify technologically challenging features, e.g., Is it new? Have others used it? For how long?

- Failure to adhere to the specifications results in feature creep, a common and critical problem that occurs when specifications are poorly written or don't exist. After this phase the addition or change of any specification will result in delays and increased costs

Specification to build a robot

- ❖ cost \$6.25
- ❖ two motors and rubber tires
- ❖ tank steering
- ❖ traces your logo but avoids the walls
- ❖ it has a plastic body with your logo on two sides.
- ❖ a domed top and round body with a 10-cm diameter

Possible question that may help to explore Specification

1. Is the specification measurable?
2. How will the specification be measured?
3. Are there formal tests that can be used?
4. How will the customer see that the design specifications will be met?
5. How many ways can the specifications be interpreted?
6. What are the failure criteria?
7. Are the numerical values and ranges clear, with units?
8. What are the tolerances?
9. How will acceptance criteria be measured? Safety, usability, etc.
10. How can we manage feature creep?
11. Are there standard tests available for a specification?

Standard and codes

Standards and Codes

ISO- International organization for standardization; ISI- Indian standard institute;

ASTM- American society for testing and materials

ANSI- American national standard institute

Standards and Codes

Standards

- A set of technical definitions, guidelines or instructions.
- Generally involves “How to” instructions for designers, engineers and manufacturers.
- Standards serve as a common language to defining language and establishing safety criteria.
- Examples : **ASTM** International standards, **ISO**, **ISI**

ISI MARK



ASTM INTERNATIONAL



Why Standard is required?

- Standards establish engineering or technical requirements for products, practices, methods or operations.
 - == Standards build confidence in customer.
 - == Reduces production cost since requirements are standardized.



Codes

Standards become Codes when:

- It is adopted by government or enforced by law or incorporated in some business or service contract.

If standards are adopted by government then standards become code of law (mandatory) **otherwise** it serves as a general guideline.

Example: **Indian Standard (IS) codes** - Used by mainly civil engineers in design and construction of buildings, dams, roads, bridges etc.

Labour codes

The codes on wages

The codes on unemployment

Example:

Bussiness code -

Code of Ethics

- Obey the law
- Take care of our members
- Take care of our employees
- Respect our suppliers

Goolge's Code of Conduct prohibits discrimination based:

- Human race
- Color
- Religion
- National origin
- Gender identity
- Age
- Disability

Some important codes



Do you know Engineering Standards?



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Standards	Description
ISO	International Organization for Standardization
ANSI	American National Standards Institute
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
ASTM	American Society for Testing and Materials
AWS	American Welding Society
EN	European Standards/European Norm
IACS	International Association of Classification Societies (for Marine safety)

Design Features

Aesthetic Design

(The power of beauty in design)

- Aesthetic design – Focus on making a design of pleasing qualities.
- Humans like **pretty and shiny design**; they desire it much more than functional one.
- Aesthetically pleasing design **satisfies our senses**.
- 4 Main Categories of aesthetic design:
 - 1. Vision:** Visual aesthetics
Color, Shape, Pattern, Line, Texture, Visual weight
 - 2. Hearing:** Sound aesthetic
Loudness, Pitch, Beat, Repetition, Melody, Pattern and Noise.
 - 3. Touch:** Material aesthetic
Shape, Weight, Comfort, Temperature, Sharpness
 - 4. Taste and Smell:** (Especially in food industry)
Sweetness, Sourness

Why aesthetic design matters?

- Beautiful products/objects are perceived as easier to use and more valuable than ugly ones. As well close bonding/attachment with design
- Aesthetically pleasing designs are bringing up positive attitude in the users. It makes them care more about the product. Aesthetic design makes people more loyal of the brand and tolerant toward mistakes or failures.

❖ *Aesthetics and usability in balance*

Smart phones that look and feel good, but also have the desired usability

❖ *Aesthetics over usability*

- ✓ Some times products have dominating aesthetics that are not supported by good usability and ergonomics. This is mostly visible in the fashion industry.
- ✓ Shoes made to look nice and attractive, while at the same time destroying the feet of the user

❖ *Usability over aesthetics*

Equipment designed for emergency situations, where people cognition is compromised

Design for Manufacturing or Production (DFM)

- Design for Manufacturing (DFM) is the process of designing parts, components or products for ease of manufacturing with an end goal of making a better product at a lower cost.
- This is done by simplifying, optimizing and refining the product design.

□ Four principles are examined during a DFM:

❖ *Process*

The manufacturing process chosen must be the correct one for the part or product.

❖ *Design*

Design is essential. The actual drawing of the part or product has to conform to good manufacturing principles for the manufacturing process you've chosen.

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❖ *Material*

It's important to select the correct material for your part/product

Some material properties to consider during DFM include:

- Mechanical properties - How strong does the material need to be?
- Optical properties - Does the material to be reflective or transparent?
- Thermal properties - How heat resistant does it need to be?
- Color - What color does the part need to be?
- Electrical properties - Does the material need to act as a dielectric (act as an insulator rather than a conductor)?
- Flammability - How flame/burn resistant does the material need to be?

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❖ *Environment*

Your part/product must be designed to withstand the environment it will be subjected to

❖ *Compliance/Testing*

All products must comply with safety and quality standards

DFM Guidelines

❖ *Minimize total number of parts*

Eliminating parts results in great savings. A part that is eliminated costs nothing to make, assemble, move, store, clean, inspect, rework, or service. A part is a good candidate for elimination if there is no need for relative motion, no need for subsequent adjustment between parts, and no need for materials to be different

❖ *Standardize components*

Costs are minimized and quality is enhanced when standard commercially available components are used in design. The benefits also occur when a company standardizes on a minimum number of part designs (sizes, materials, processes) that are produced internally in its factories

❖ *Use common parts across product lines*

It is good business sense to use parts in more than one product. Specify the same materials, parts, and subassemblies in each product as much as possible.

❖ *Standardize design features*

Standardizing on design features like drilled hole sizes, screw thread types, and bend radii minimizes the number of tools that must be maintained in the tool room. This reduces manufacturing overhead cost

❖ *Design parts to be multifunctional*

A good way to minimize part count is to design such that parts can fulfill more than one function, leading to integral architecture

❖ *Minimize secondary and finishing operations*

Minimize secondary operations such as heat treatment, machining, and joining and avoid finishing operations such as deburring, painting, plating, and polishing. Machine a surface only when the functionality requires it or if it is needed for aesthetic purposes.

❖ *Avoid excessively tight tolerances*

Tolerances must be set with great care. Specifying tolerances that are tighter than needed results in increased cost

Design for Standardization

- The benefits of standardization occur in four areas: cost reduction, quality improvement, production flexibility

❖ *Cost reduction*

✓ *Purchasing cost*

Standardization of parts and the subsequent reduction in part numbers will result in large savings in procurement costs in outsourcing because parts will be bought in larger quantities

✓ *Reduce costs through raw material standardization*

Cost for in-house production of parts can be reduced if raw materials can be standardized to a single size of bar stock, tubing, and sheet metal

✓ *Feature standardization*

Standardizing on design features like drilled hole sizes, screw thread types, and bend radii minimizes the number of tools that must be maintained in the tool room. This reduces manufacturing overhead cost

Continue...

❖ *Quality improvement*

✓ *Product quality*

Having fewer parts of a given type greatly reduces the chance of using the wrong part in an assembly

✓ *Prequalification of parts*

The use of standard parts means that there is much greater cumulative experience with using the particular part. This means that standard parts can be prequalified for use in a new product without the requirement for extensive testing

✓ *Supplier reduction means improved quality*

Standardization of parts means there will be fewer outside suppliers of parts. Those suppliers remaining should be those with a record of producing quality parts

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Production flexibility

✓ *Material logistics*

The flow of parts within the plant will be easier with fewer parts to order, receive, stock, issue, assemble, test, and reorder

✓ *Parts availability*

Fewer part types used in greater volume will mean less chance of running out of parts and delaying production

✓ *Quicker supplier deliveries*

Standardization of parts and materials should speed up deliveries. Suppliers will have the standard tools and materials in their inventory

Design for safety

- A safe product is one that does not cause injury or property loss
- Achieving safety is no accident. It comes from a conscious focus on safety during design, and in knowing and following some basic rules

There are three aspects to design for safety:

- ❖ Make the product safe; that is, design all hazards out of the product.
- ❖ If it is not possible to make the product inherently safe, then design in protective devices like guards, automatic cutoff switches, and pressure-relief valves, to mitigate the hazard.
- ❖ If step 2 cannot remove all hazards, then warn the user of the product with appropriate warnings like labels, flashing lights, and loud sounds.
- ❖ Provide training and protective clothing or devices (glasses, ear mufflers) to the user or operator of the equipment

Design for safety

A fail-safe design

- A fail-safe design seeks to ensure that a failure will either not affect the product or change it to a state in which no injury or damage will occur.

There are three variants of fail-safe designs

❖ *Fail-passive design*

When a failure occurs, the system is reduced to its lowest-energy state, and the product will not operate until corrective action is taken. A circuit breaker is an example of a fail-passive device

❖ *Fail-active design*

When failure occurs, the system remains energized and in a safe operating mode. A redundant system kept on standby is an example

❖ *Fail-operational design*

The design is such that the device continues to provide its critical function even though a part has failed. A valve that is designed so that it will remain in the open position if it fails is an example

Design to cost

- Design to cost, also called target costing, is the approach in which a target value, (sometimes called “should-cost” data), for the cost of a product is established at the beginning of a product development project.
- All design decisions are examined for their impact on keeping below the target cost

The steps in accomplishing design to cost are

❖ *Establish a realistic and reliable target cost*

The target cost is the difference between a realistic estimate of what the customer will pay for the product when developed minus the expected profit

❖ *Divide the target cost into subunits*

- ✓ division according to competitors' component costs, as determined from dissection of competitor products
- ✓ on the basis of estimates of what the customer is willing to pay for various functions and features of the product

❖ *Oversight of compliance with cost targets*

Major difference in the design to cost approach is that the cost projections will be evaluated after each design phase as well as before going into production

Order of Magnitude Estimates

At the very early stage of product development where the market for a new product is being studied, comparison is usually made with similar products already on the market. This gives bounds on the expected selling price.

Products can be divided roughly into three categories

- ✓ Large functional products—automobile, front-end loader, tractor
- ✓ Mechanical/electrical—small appliances and electrical equipment
- ✓ Precision products—cameras, electronic test equipment

Products in each category cost roughly the same on a weight basis

□ A slightly more sophisticated method is to estimate cost on the basis of the percentage of the share of the total cost that is due to materials cost. For example, about 70 percent of the cost of an automobile is material cost, about 50 percent for a diesel engine, about 25 percent for electrical instruments, and about 7 percent for China dinnerware.

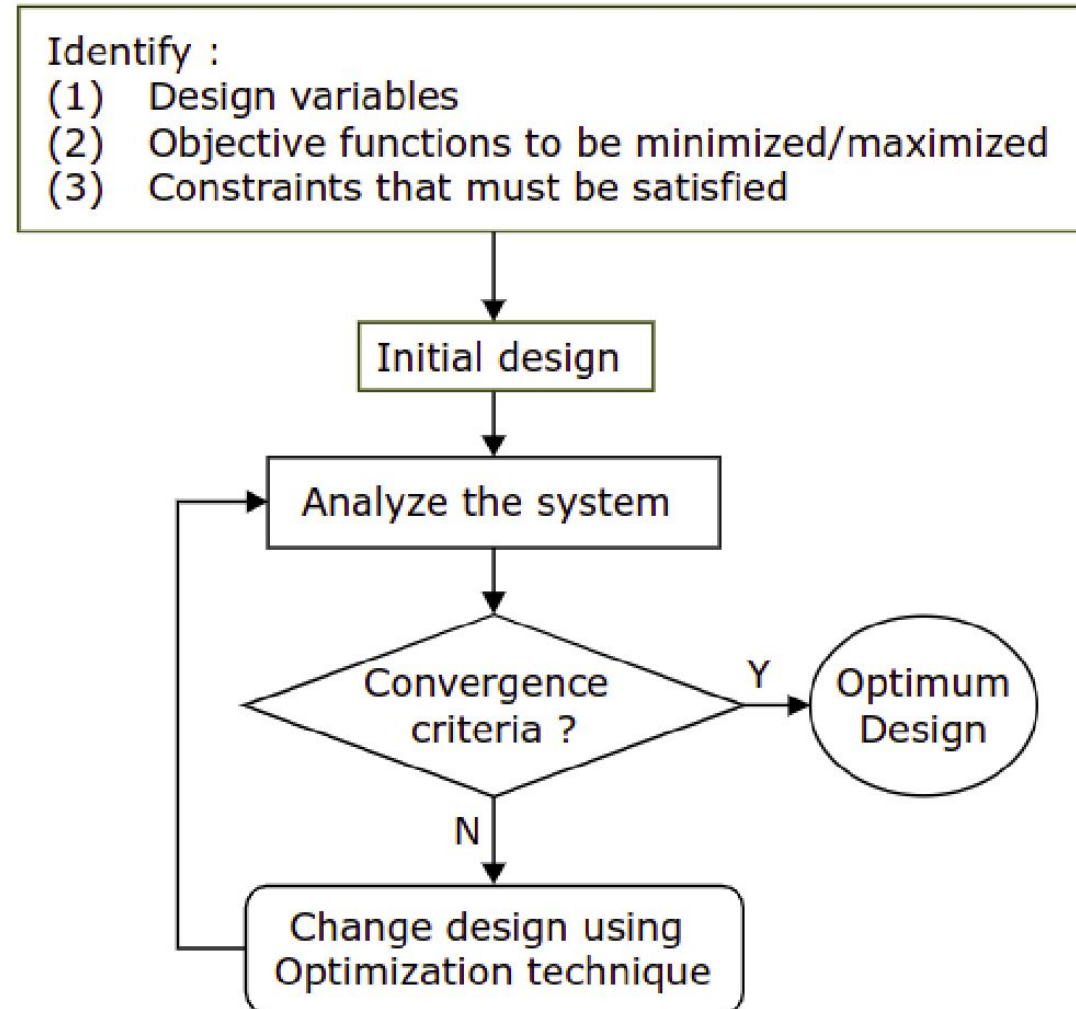
□ **What is the total cost of a diesel engine that weighs 300 lb? The engine is made from ductile iron that costs \$2/lb. The material cost share for the engine is 0.5**

$$\text{Cost} = (300 \$2) / 0.5 = \$1200$$

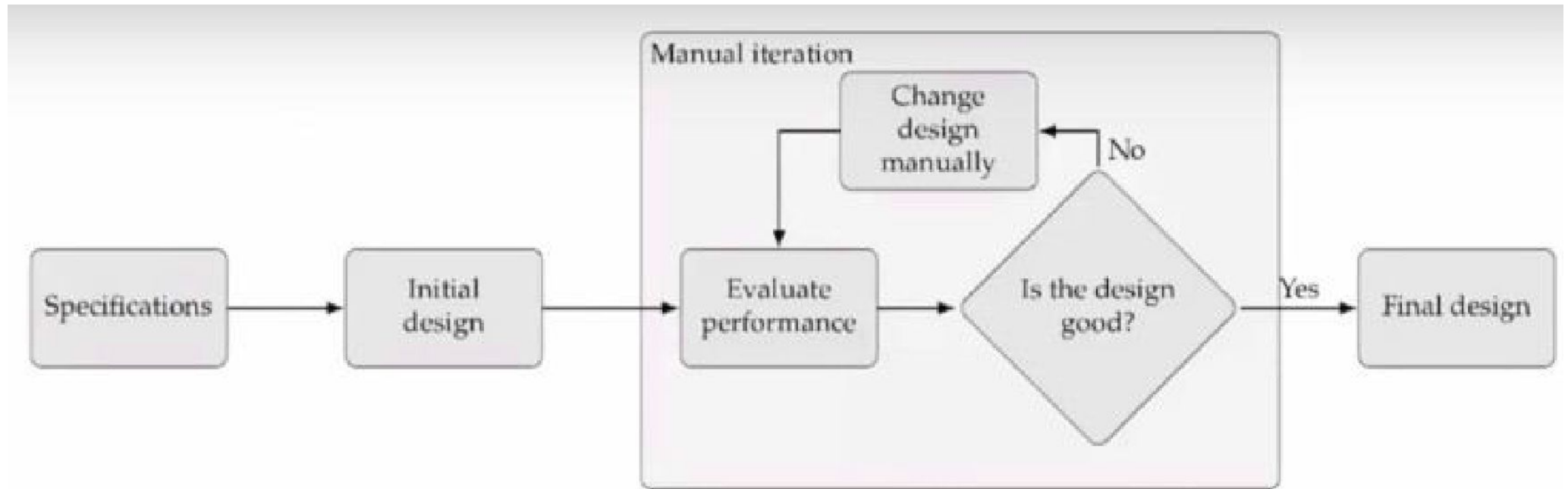
Optimum Design

Optimum Design

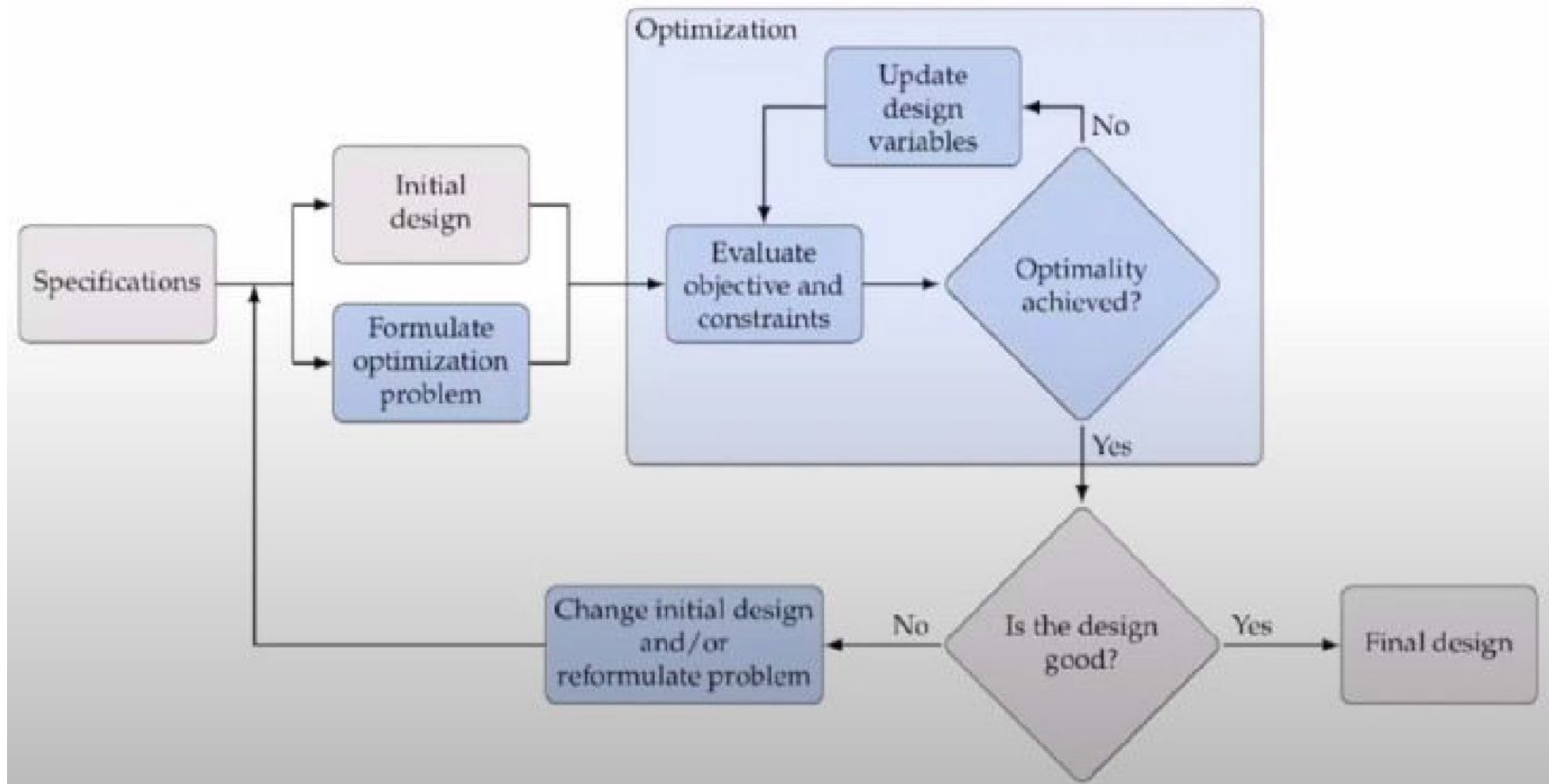
- Optimum design is a best possible design satisfying certain criteria among many feasible or adequate design
- Mathematical, it is similar to finding a maximum or minimum for a function



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Design Optimization is at work everywhere: manufacturing, transportation, logistics, financial services, utilities, energy, telecommunications, government, defense and retail

Engineering Applications

- Vehicle design and analysis

- Shape optimization of Aircraft structures

- Heat transfer analysis in CVD(chemical vap. deposition) reactors

- Size and topology optimization of mechanical components

Medical Applications

- Algorithms for new drug development

- Protein structure assessment

- DNA sequence mapping

Environmental protection

- Efficient soil and water resource utilization

- Land and natural resource allocation

Management Mathematics

- Supply Chain Management,

- Distribution Management

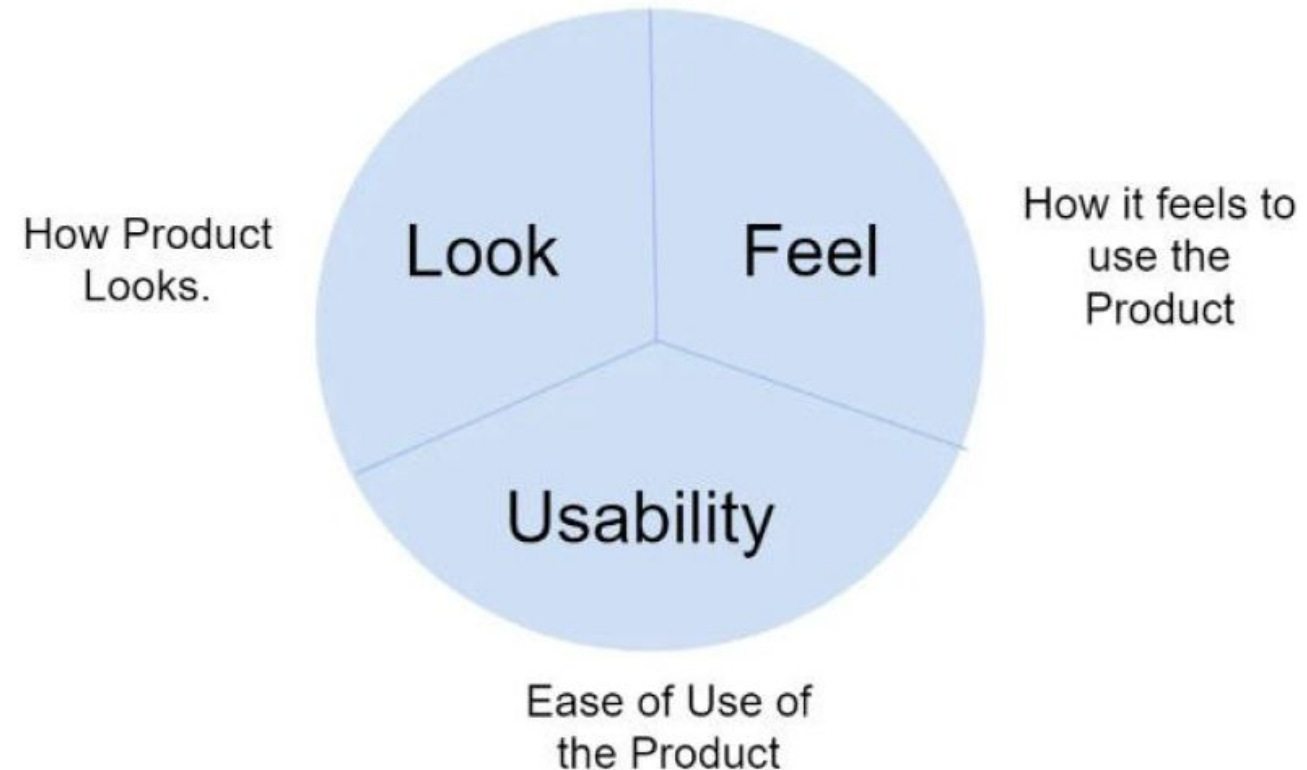
- Production Scheduling

Usability

Design for Usability (or) Design for Ease of Use

- Usability is a measure of how well a user can use a product/design effectively, efficiently and satisfactorily.
- Usability is a component of user experience (UX).
- A design's usability depends on how well its features accommodate users' needs and contexts.

User Experience (UX)



Elements to consider as a Designer while focussing on Usability

1. Effectiveness—It supports users in completing actions accurately.

2. Efficiency—Users can perform tasks quickly through the easiest process.

3. Engagement—Users find it pleasant to use and appropriate for its industry/topic.

4. Error Tolerance—It supports a range of user actions and only shows an error in genuine erroneous situations. Achieved by the number or type of severity of common errors users make, as well as how easily users can recover from those errors.

5. Ease of Learning—New users can accomplish goals easily and even more easily on future visits.



Usability Testing

- A process of evaluating a product or service by testing it with representative users.
- **Typically, during a test,**
Participants will try to complete typical tasks while **Observers** watch, listen and takes notes.



Goal

- Identify any usability problem
- Collect qualitative and quantitative data
- Determine participant satisfaction's

Benefits of usability Testing

During usability testing, you (the host) will

- Learn if participants are able to complete specified tasks successfully
- Identify how long it takes to complete specified tasks
- Find out how satisfied participants are with your product
- Identify changes required to improve user performance and satisfaction
- Analyze the performance to see if it meets your usability objectives

Factors affecting testing cost

- Type of testing performed
- Size of the team assembled for testing
- Number of participants for testing
- Number of days you will be testing

Elements to consider when budgeting for usability testing:

- Time
- Recruiting Costs
- Participant Compensation
- Rental Costs

Customer Co-creation

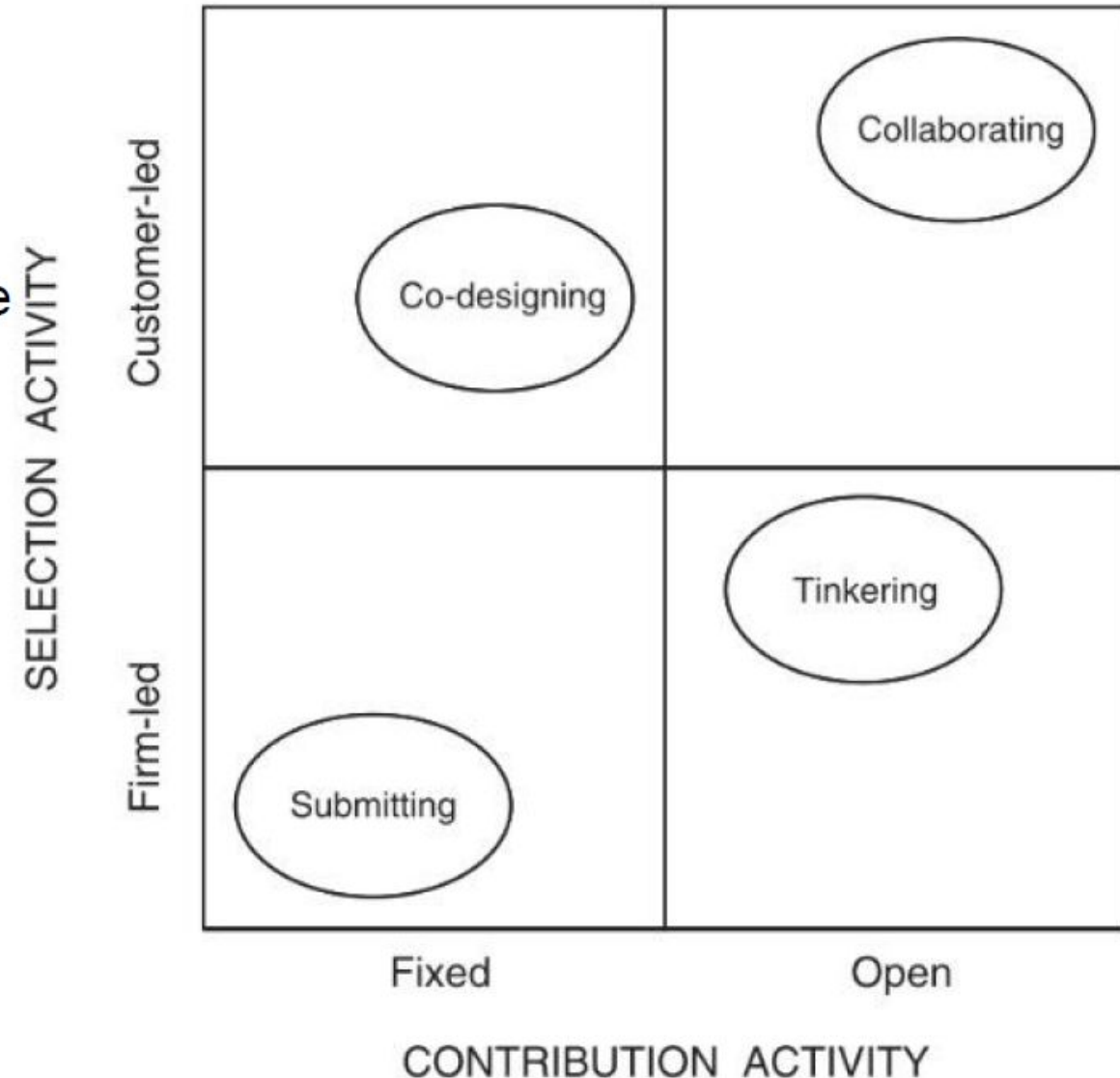
Co-creation refers to **inviting stakeholders (usually customers or employees) to participate in a design or problem-solving process** to produce a mutually valued outcome.

Collaborating: A process in which customers have the power to collectively develop and improve a new product's core components and underlying structure.

Tinkering: A process in which customers make modifications to a commercially available product and some of these modifications are incorporated into subsequent product releases.

Co-designing: A process in which a relatively small group of customers provides a firm with most of its new product content or designs.

Submitting: a process in which customers directly communicate ideas for new product offerings to a firm



- **Unilever**

Through its Open Innovation platform, launched in 2010, Unilever presents specific challenges to the public, encouraging individuals to submit responses for potential adoption by the company.

- **BMW : Co-creation lab**

- **IKEA**

IKEA's co-creation platform focuses on four specific areas:

Asking customers for product idea suggestions

Running IKEA Bootcamps to work with entrepreneurs

Collaborating with university students on product solutions

Connecting with innovation labs around the world